order in which we run them is a result of time constraints and the importance we assign to those tasks.

A common example of an environment where tasks are executed by priority is an emergency room, where patients are visited not according to the order they arrived but instead depending on how urgent their conditions are. If we move closer to our IT domain, there are many tools and systems that has the same behavior. Think, for instance, about your operating system scheduler. Or maybe you are using a mobile app to take a to-do list.

2.2.1 Priority in practice: bug tracking

The example I'd like to use in this chapter, though, is a bug-tracking suite. You are probably already familiar with such a tool: when you work in teams, you need a way to track bugs and tasks, so that no two people work on the same issue, duplicating effort, and also issues are tackled in the right order (whatever that is, depending on your business model).

To simplify our example, let's restrict to the case of a bug tracking tool where each bug is associated with a priority, expressed as the number of days within which it needs to be solved (lower numbers mean higher priority). Also, let's assume that bugs are independent, i.e. there is no bug that requires solving another bug as a prerequisite.

For our example, let's consider the following list of bugs (in sparse order) for a single page web application.